Remanent Magnetism of Carboniferous Limestone

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(Received 1961 March 27)*

Summary

Paleomagnetic data are lacking on Carboniferous limestones of England and during an investigation of iron ores the author found specimens of these limestones that could be measured. An average declination of S 41 W and inclination of minus 29 degrees was determined for the remanent magnetization. This gives a pole position of north 40 degrees latitude and 122 degrees east longitude. Intensity values of remanent magnetization ranged upwards to $27{\cdot}5 \times 10^{-6}$ emu/cm^3.

1. Introduction

During the course of a study of the magnetism of rocks of ore deposits and associated areas, the author measured the remanent magnetism of a series of specimens of British hematitic iron ore and contiguous Carboniferous limestone. This paper pertains to the remanent magnetism of the Carboniferous limestone; results on the hematites will be forthcoming.

The specimens were collected during the winter of 1959–60, from an area immediately southwest of the Lake District of England. Specifically they came from sites located at N 54°, W 3°. These locations are on either the Barrow-in-Furness or Keswick one inch maps of the Ordnance Survey of Great Britain.

The field orientation of the specimens was determined with the aid of a Brunton compass after it had been determined that there were no local anomalous magnetic effects that would effect the orientation of the compass needle. From the oriented specimens oriented discs were obtained that measured three-fourths of an inch in diameter and five-eights of an inch in length. A total of 52 disks were prepared from 19 specimens and of this total 22 disks could be measured from 14 specimens. Corrections for geological dip were made by coring normal to bedding surfaces of specimens collected at one site where dips of 8° were measured. At other sites the beds were considered to be flat lying.

The specimens measured are fine- to medium-grained, grey to reddish brown, limestone of Carboniferous age (Smith 1919). The rocks have been locally altered and even more locally have been metasomatically replaced and openings filled by hematite.

2. Experimental and calculated results

Magnetic measurements were made on the prepared disks by the methods described by Collinson, & others (1957). For the size of specimens used and for the

* Received in original form 1960 December 28.

conditions of instrumentation, a lower limit of reliable measurement is considered to be $2 \cdot 10^{-7}$ emu/cm^3. Results are given in Table 1 and are plotted in Figures 1 to 3. In the table, under the heading "declination," the azimuth of the north seeking pole is noted under "N pole." Its inclination is given and a minus sign means that the angle is measured above the horizontal. To determine the general direction, an average was taken of the southwest data and a second one of the northeast data by the use of direction cosines. In this treatment, individual disks are weighted equally in the overall result and the final average reading is influenced to a greater degree by those sites and samples upon which more readings were made. According to this method of calculation, an average declination of S 41 W and an inclination of $-29°$ is obtained from the southwest data. Similarly for the northeast data a value of N 34 E is obtained for the declination and the value for inclination is $+53°$. A pole position for the southwest data is calculated at N 40°, E 122°. One is not calculated for the northeast data because of its lack of significance.

Table 1

Specific Gravity	Intensity 10^{-6} emu per cm^3	Intensity 10^{-6} emu per gm	Susceptibility 10^{-6} emu per cm^3	Susceptibility 10^{-6} emu per gm	Orientation Declination N Pole	Orientation Inclination (− up deg.)
2·39	0·8	·3	1	1	S59W	−51
2·56	1·4	·6	1	1	S72W	−44
2·48	21·2	8·6	3	1·2	S24W	−30
2·54	9·1	3·6	7·5	3·0	S35W	−35
2·59	18·6	7·2	4	1·5	S25W	−28
2·62	5·9	2·3	5	1·9	S26W	−23
2·54	7·4	2·9	4	1·6	S15W	−3
2·61	15·6	6·0	5·5	2·1	S10W	−11
2·62	27·5	10·5	5·5	2·1	S41W	−38
2·64	6·4	2·4	8·5	3·2	S44W	−31
2·64	20·7	7·8	4·5	1·7	S42W	−38
2·62	11·6	4·4	5·5	2·1	S48W	−34
2·53	24·0	9·5	5·5	2·2	S24W	−33
2·61	5·8	2·4	5·5	2·1	S45W	−29
2·56	7·7	3·0	7	2·7	N25E	+56
2·59	1·4	·5	1·5	1	N26E	+57
2·63	1·4	·5	1·5	1	N28E	+46
2·64	1·2	·4	1	1	N49E	+54
2·51	0·9	·4	1	1	S65W	−19
2·66	0·4	·2	1	1	S66W	−32
2·59	0·9	·3	1	1	S60W	−4
2·58	1	·4	1	1	S66W	−17

A measure of the uncertainties of the averages for the southwest data is given by applying Fisher's (1953) best estimate of the precision parameter κ. Calculations give a value for κ of 18·9. Determinations of cone of confidence for α_{95} give values of 7·6°.

3. Discussion

The discussion of the magnetic data as set forth in this paper is limited to simple presentation and conclusion. The reader is referred to the excellent review

article by Cox & Doell (1960) on the subject of paleomagnetism for a fuller treatment and for a discussion of pertinent theoretical aspects. The results of the magnetic measurements as given in Table 1 show considerable uniformity. Specific gravity values range from 2·39 to 2·66, and all are below the theoretical 2·72 value for calcite. Intensities of remanent magnetism range from values too weak to measure up to $27{\cdot}4 \times 10^{-6}$ emu/cm^3. A comparison of specific gravity with magnetic intensity as plotted in Figure 1 shows only a general correlation. This suggests that for ranges of low concentrations, variable amounts of ferro-magnetic material

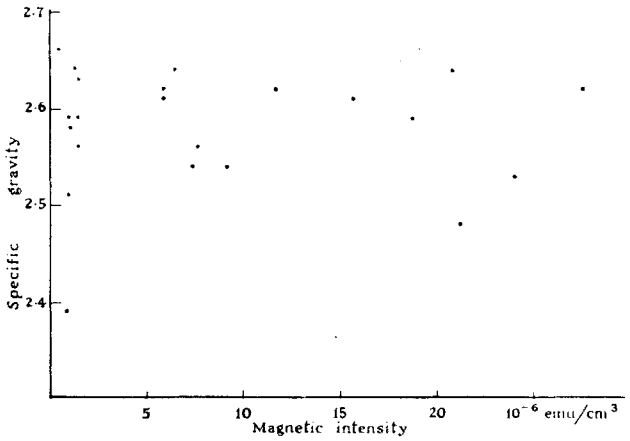


FIG. 1.—Specific gravity vs. magnetic intensity.

will have greater effects on magnetic properties than on specific gravity. Also for these ranges of concentration, individual characteristics, shape, grain size, composition, etc., will have a greater influence on intensity.

The results of susceptibility measurements given in Table 1 range from values too small to measure up to $8{\cdot}5 \times 10^{-6}$ emu/cm^3. These are related to values of intensity of remanent magnetism in Figure 2. These two variables are broadly related for the rocks studied.

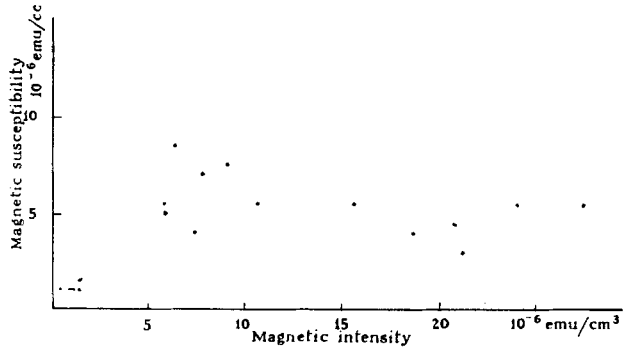


FIG. 2.—Magnetic susceptibility vs. magnetic intensity.

The orientation of the remanent magnetization is given in Table 1 and plotted in Figure 3. Readings taken on disks from the same specimen are connected together by a full line. Two distinct and separate groupings of orientatious

are obtained, a minor one to the northeast with a downward plunge, and a major one to the southwest with an upward plunge. The major direction is considered a normally magnetized direction resulting from a chemical magnetization associated with hematite formation. That it is a result of chemical magnetization is suggested by geological evidence which includes a complete lack of post-depositional thermal effects in the measured and associated rocks and also includes considerations of the overall geological environment of these rocks during and since formation.

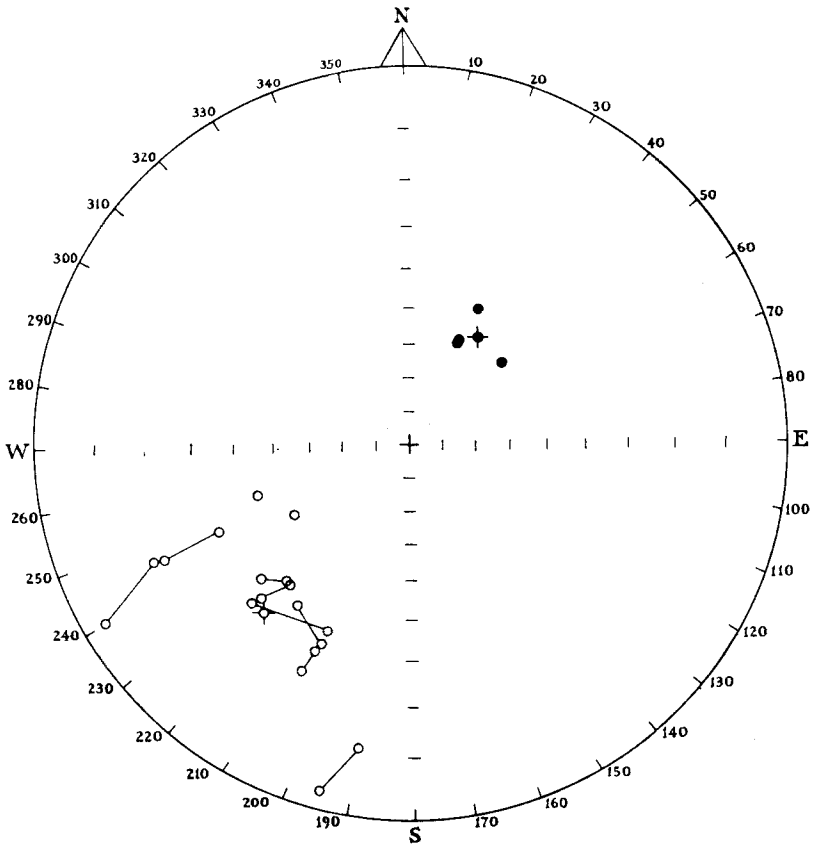


FIG. 3.—Direction of remanent magnetism.

Because of the distribution and position of the magnetic orientations, the magnetization is considered to have essentially no component of isothermal remanent magnetization. The minor direction is considered a reversed direction, the result of some self-reversal mechanism or of a magnetization during a reversed Earth's field. Everitt & Belshé (1960) concluded this latter case for some of their rocks and further concluded that the Earth's field has undergone a series of reversals during early Carboniferous times and to have finally remained reversed for a long period of time.

The magnetic orientations measured agree in part with data of Nairn (1960). The calculated paleomagnetic pole positions agree favourably with those published by Collinson & Runcorn (1960) and by Cox & Doell (1960). This agreement is not exact and of course some deviation occurs.

Acknowledgments

The author wishes to express his sincere thanks to Professor S. K. Runcorn for his help and for the facilities that he furnished. Acknowledgments are also gratefully given to the other members of the rock magnetism group at King's College, Newcastle-upon-Tyne, and especially to Dr A. E. M. Nairn and D. W. Collinson. The samples were prepared and magnetism measurements made at King's College, Newcastle-upon-Tyne.

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1961 *March* 22.

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